# Rajveer Gupta

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**Portfolio: live demos, project case studies and the IEEE paper**

## Education

**Indian Institute of Information Technology, Nagpur** Aug 2024 – Jun 2028 (expected)
***B.Tech, Computer Science & Engineering — CGPA: 9.08/10*** *Nagpur, India*
**Coursework:** DSA, OOP (AA), Software Engineering (AA), Operating Systems (AB), Programming Language Design (AB), DBMS

## Publication

A. Gupta, A. Singh, **R. Gupta**, P. Jain, T. Patidar, "Fault Tolerant Industrial Vision Framework Through Semantic-Guided Cross-View Temporal Reconstruction and Restoration," *IEEE IRAI 2026*, Melbourne (accepted and presented) | **Paper**

## Experience

**Research Intern, Computer Vision — IIIT Nagpur (Advisor: Dr. Nishat Afshan Ansari)** 2026
*HocVid · PyTorch, OpenCV, NoPoSplat (3D Gaussian Splatting), CoTracker2, SAM, DiffuEraser, MiDaS, Mamba*
**Overview:** co-developed HocVid, which detects a dead camera in a **3**-camera network and rebuilds the lost view from overlapping cameras and the camera's last **16** frames.

- Detected camera faults with a degradation score (frequency anomalies, cross-camera SSIM, hardware flags) that labels cameras healthy, degraded or dead; a 16-frame **hysteresis** window stops flicker: **89.4% accuracy**.
- Calibrated cameras with SIFT + RANSAC (fundamental matrix, intrinsic and extrinsic parameters); rebuilt surviving views as **pose-free 3D Gaussians** (NoPoSplat) from the failed camera's stored pose, worth **1.85 dB**.
- Tracked SAM object centroids across the dead camera's last 16 frames with **CoTracker2**; low-visibility tracks mark dead zones for video diffusion (DiffuEraser), adding **1.18 dB**.
- Restored frames by joining frozen backbones with small trained adapters and rank-16 LoRA (3.7M new parameters), then super-resolution and frame interpolation; a quality gate rolls back hallucinated frames.
- Evaluated under injected physical faults (blur, lens dust, thermal noise, signal loss): **30.02 dB PSNR** on Wildtrack, +3.65 dB over the best naive cascade; **32.42 dB / 0.953 SSIM** on GoPro deblurring, **+2.37 dB** over MoCE-IR.

## Projects

**Resolv — Multilingual Entity Resolution over 11.7M Records (Amazon ML Challenge 2026)** 2026
*GitHub / Live / Demo — Team lead of 4 · **AIR 471 of 32K teams** · PyTorch, multilingual-e5, Qwen3, LightGBM*

- **Diagnosed a train–test distribution shift** (**39.8%** vs **26.0%** unmatched records) behind a **0.963** → **0.94** drop from validation to leaderboard; simulated it in training and refit every model.
- Combined IDF-weighted keys from **11** families with a multilingual-e5 bi-encoder fine-tuned with InfoNCE: **99.86%** pair recall; a LightGBM re-ranker cut **86.7** candidates per business to **5.58**.
- Stacked **107** hand-built features with a cross-encoder and a fine-tuned **Qwen3-Reranker-0.6B** on uncertain pairs, lifting leaderboard F0.5 from **0.94** to **0.986** (**0.998** validation precision).

**Memora — Memory-Care App for Alzheimer's Patients** 2025 – 2026
*GitHub / Live / Results / Case study — Python, Flask, ONNX, PyTorch, RoBERTa · Sole developer*
**Overview:** tells a patient who is in front of them in a loved one's voice and finds medicines; rebuilt solo from ARKA.

- Raised LFW face recognition from **87.1%** to **98.9%** (95% CI **98.2–99.5**) and cut latency from **395** to **21 ms** by fixing detection and alignment first (YuNet + SFace).
- Kept stranger false matches at **1.3%** with **200** people enrolled (fixed threshold: **17.4%**) using a threshold scaled to gallery size, with Platt-calibrated confidence (ECE **0.009**).

**ReturnGuard — Multi-Agent Return-Fraud Detection for E-commerce** 2026
*GitHub / Live / Demo — Python, LangGraph, Gemini, PyTorch, DINOv2, Pydantic, Flask, SQLite, GitHub Actions*
**Overview:** checks at the doorstep that a returned product is the unit that was delivered, before the refund.

- Fixed the vision signal: **65–77%** of DINOv2 patches were square padding, so object-only pooling cut a false "similar" score from **0.77** to **0.38**; a colour check catches same-model colour swaps.
- Measured with confidence intervals: **100%** precision, **93.3%** recall and **0.986** F0.5 on **48** labeled pairs (v1: **90.9%** precision); **97.4%** precision at **100%** recall across **36** photo-distortion conditions.

## Achievements

- **Led the team** in **Amazon ML Challenge 2026** to **AIR 471 of 32K teams**.
- **Amazon ML Summer School 2026**, selected in the **top 3,000 of 134,000+** applicants.
- **Led the team** in **Microsoft Imagine Cup 2025** to the **MVP Round** (**60K+** teams); managed the full ML pipeline.
- **Smart India Hackathon 2025**, **National Semi-Finalist** (AyurSutra) among **100,000+** participants.
- **Best Innovation Award**, IIT Hyderabad (Memora) • IEEE IES GenAI Hackathon Finalist.

## Skills

**Computer Vision:** OpenCV, SIFT/RANSAC, 3D Gaussian Splatting, CoTracker2, SAM, DINOv2, diffusion, ONNX
**Deep Learning & NLP:** PyTorch, TensorFlow, Transformers, multilingual-e5, Qwen3, Mamba, LoRA/QLoRA
**Programming & Tools:** Python, C++, C, Java, SQL; Flask, Docker, Git, Linux
**Machine Learning & Data Analysis:** NumPy, pandas, Polars, scikit-learn, LightGBM, MLflow